



BASIC CALCULUS

ACADEMIC

Grade 11/12

Curriculum Guide

BASIC CALCULUS
Prerequisite: Pre-Calculus

Course Description:

This course equips learners with advanced mathematical skills and conceptual understanding essential for success in higher level Calculus and related fields. Focusing on derivatives and tangents, rules of differentiation, applications of derivatives, antiderivatives, and definite integrals, the course develops deep analytical thinking, rigorous problem-solving, and sound mathematical reasoning. Using technology and collaborative learning, learners investigate the behavior of functions, compute derivatives and integrals using standard rules and techniques, and explore the connections between algebraic processes and graphical interpretations. Learners construct and interpret tables, graphs, and mathematical models to illustrate limits and continuity, examine the derivative as the slope of a tangent line, and apply Calculus concepts to real-life situations in physics, business, economics, and engineering, and related fields.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT DOMAIN	CONTENT STANDARDS <i>The learners demonstrate knowledge and understanding of ...</i>	LEARNING COMPETENCIES <i>The learners ...</i>
NUMBER AND ALGEBRA (NA)	1. Limits and Continuity	1. illustrate the limit of a function using a table of values and its graphical representation, emphasizing the behavior of a function as it approaches a specific point. 2. apply limit laws to evaluate left-hand limits, right-hand limits, and limits of polynomial, rational, radical, exponential, and logarithmic functions. 3. analyze continuity at a point by determining whether a function is continuous at a given number. 4. examine the continuity of a function over an interval.

CONTENT DOMAIN	CONTENT STANDARDS <i>The learners demonstrate knowledge and understanding of ...</i>	LEARNING COMPETENCIES <i>The learners ...</i>
		5. solve problems involving continuity of functions.
	2. Derivatives and Tangents	6. define and illustrate the tangent line to the graph of a function at a given point. 7. explain the derivative as rate of change and as slope of a tangent line. 8. apply the definition to calculate the derivative of a function at a given number. 9. determine the relationship between differentiability and continuity of a function.
	3. Rules of Differentiation	10. apply differentiation rules to compute the derivatives of algebraic, exponential, and logarithmic functions. 11. calculate derivatives using the Chain Rule and implicit differentiation. 12. solve problems involving the Chain Rule and implicit differentiation of functions, including logarithmic functions.
Performance Standards <i>The learners can analyze and model real-life and mathematical situations by interpreting graphs to describe the behavior and properties of these functions and applying these insights in demonstrating continuity and limits where appropriate. They can apply their understanding of derivatives to solve problems, illustrate tangent lines, and relate derivatives to slopes. The learners also examine the relationship between continuity and differentiability and compute derivatives of algebraic, exponential, and logarithmic functions using basic differentiation rules, the Chain Rule, and implicit differentiation.</i>		
NUMBER AND ALGEBRA (NA)	4. Applications of Derivatives	13. solve problems involving rates of change and related rates. 14. solve optimization problems.
	5. Antiderivatives (Indefinite Integrals)	15. define and illustrate an antiderivative of a function. 16. explain the relationship between derivatives and antiderivatives.

CONTENT DOMAIN	CONTENT STANDARDS <i>The learners demonstrate knowledge and understanding of ...</i>	LEARNING COMPETENCIES <i>The learners ...</i>
		17. compute the general antiderivatives of polynomial, radical, rational, and exponential functions. 18. compute antiderivatives using the substitution rule. 19. solve problems involving antidifferentiation.
	6. Definite Integrals and Area	20. illustrate the definite integral as the limit of Riemann sums. 21. compute the definite integral of a function using the Fundamental Theorem of Calculus. 22. compute the definite integral of a function using the substitution rule. 23. solve problems involving areas of plane regions.
Performance Standards <i>The learners can</i> interpret and solve problems involving derivatives, related rates, and optimization in practical contexts, such as maximizing profit, minimizing cost, or modeling motion. They can also demonstrate their understanding of antiderivatives and integrals by solving real-life and mathematical problems. The learners can illustrate antiderivatives, explain the relationship between derivatives and antiderivatives, and compute antiderivatives of polynomial, radical, and exponential functions. They can apply the substitution rule to compute antiderivatives and solve related problems. The learners can interpret the definite integral as the limit of Riemann sums and apply the Fundamental Theorem of Calculus to evaluate definite integrals. They can solve problems involving areas of plane regions and apply integration in real-world contexts, such as calculating land area, material usage, and accumulated quantities.		